

Project Summary

This project focuses on analysing football match footage using computer vision and real-time frame processing. A user uploads a video, which is then split into individual frames. These frames are processed using the YOLO object detection model via Roboflow to identify players, referees, and goalkeepers. Each detection is visually labelled with bounding boxes based on class. Once all frames are processed, they are reconstructed into a complete video for seamless viewing. This system enables automated performance analysis, referee assistance, and visual feedback for tactical review.

Learning Outcomes

- Applied computer vision and deep learning
- Built a scalable full-stack system
- Experience with real-world tools like Docker, RabbitMQ, YOLO

Applications

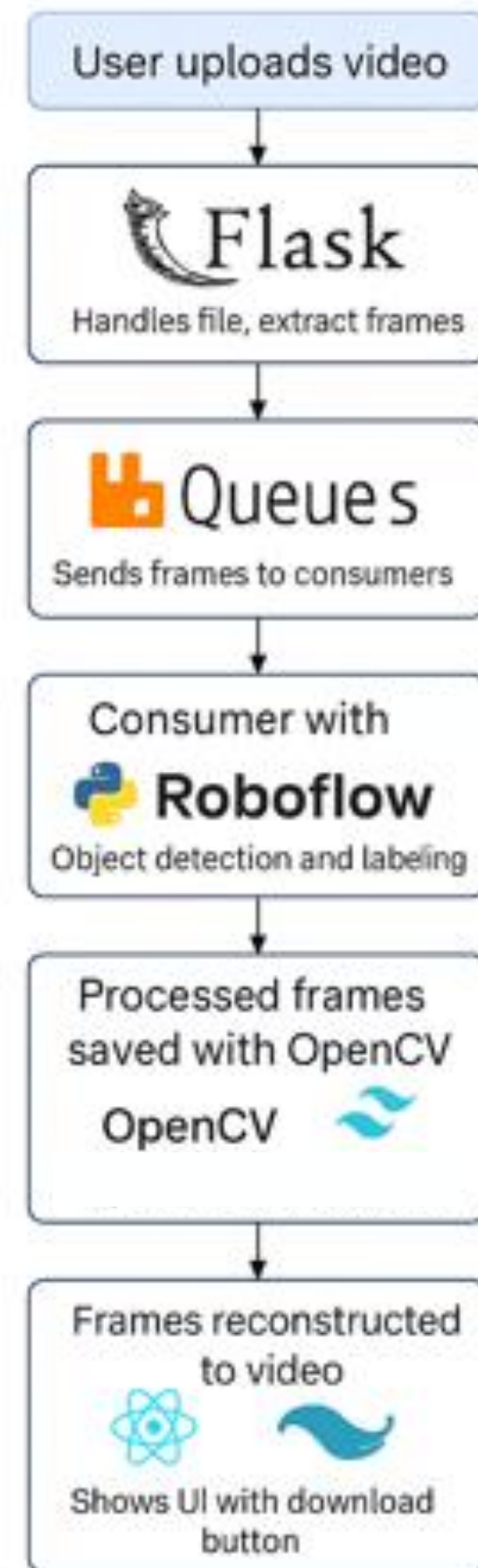
- Sports analytics
- Referee training

Repository



Architecture Diagram

Architecture Diagram



Why This Architecture Works

- ✓ **Scalable** – Multiple consumers can process frames in parallel
- ✓ **Accurate** – YOLO ensures high precision in player detection
- ✓ **Efficient** – Queue-based design avoids bottlenecks
- ✓ **User-friendly** – Output delivered in an easy-to-download video

Results & Performance

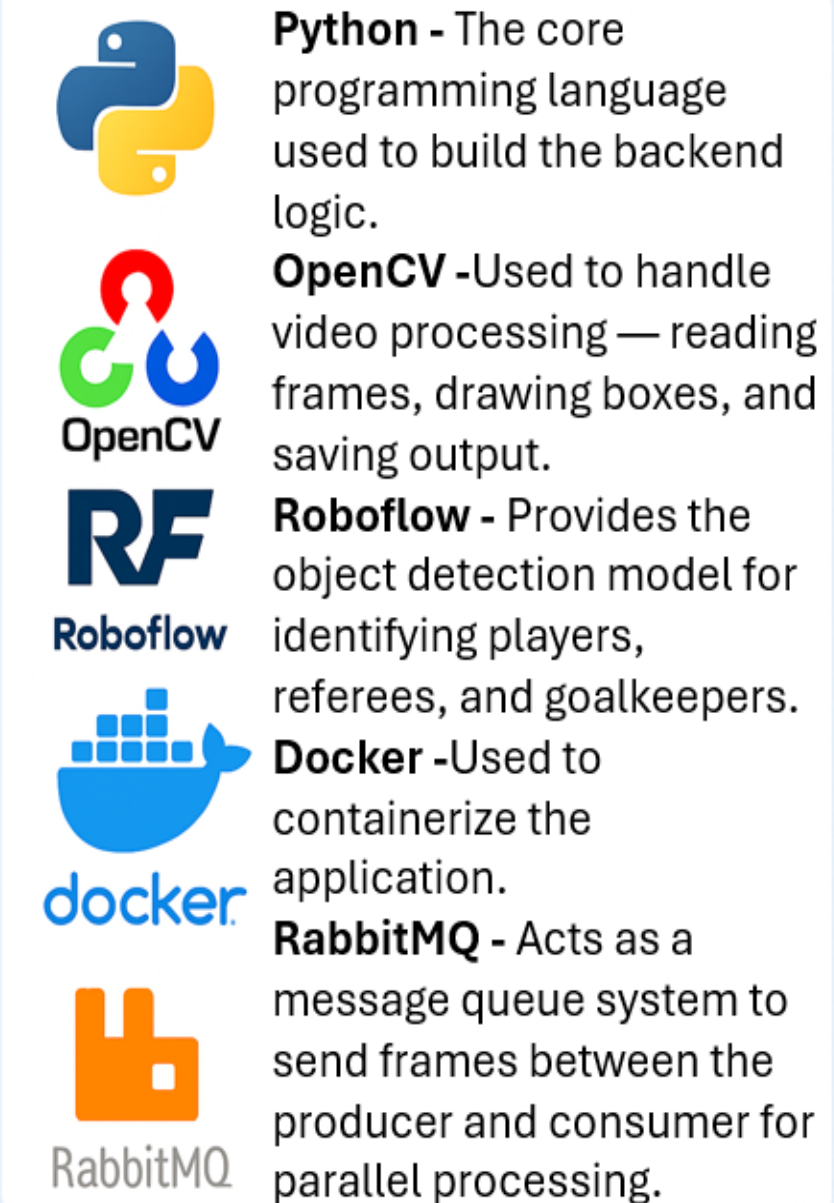
- ⚙️ **Processing Speed:**
30-second video processed in 8-10 minutes on GPU
- 🧠 **Model Accuracy:**
YOLOv8 detected:
Players: 91% precision
Referees: 85% precision
- ⬇️ **Output:**
Final video downloaded with bounding boxes and labels

What is Real Time Frame Processing?

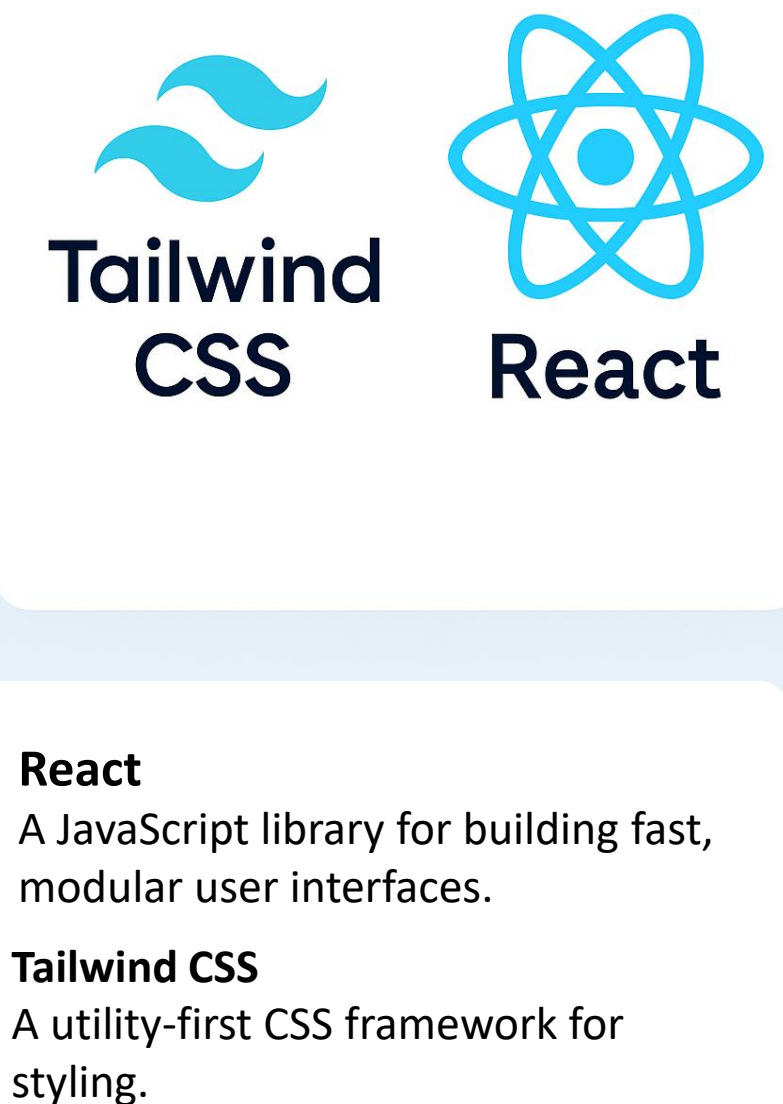
Real-time frame processing is a core technique in video analytics. It transforms raw video feeds into enhanced or analysed output with virtually no perceptible delay. This project extracts video frames and sends them through RabbitMQ, enabling parallel processing by multiple consumers. Using the YOLO model via Roboflow, it detects players, goalkeepers, and referees. Processed frames are then reconstructed into a final video, offering efficient and scalable football match analysis.



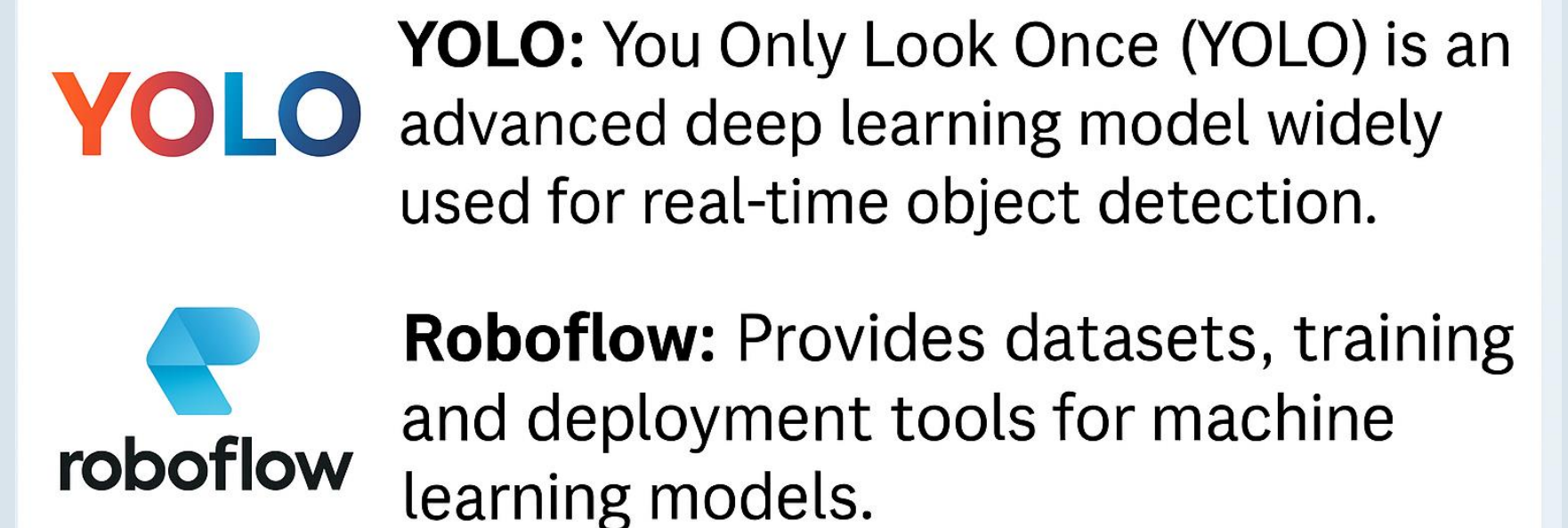
Backend & Processing



Frontend



Technical Details



Main Menu

